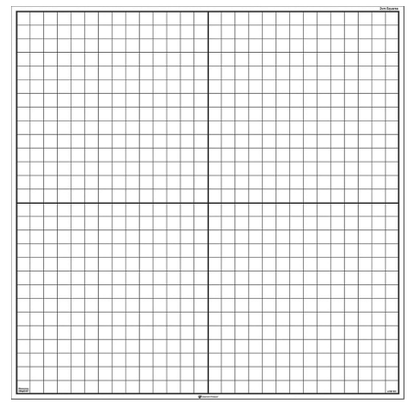
UNIVERSIDAD DEL VALLE DE ATEMAJAC
PREPARATORIA
2do examen parcial CÁLCULO

Nombre _____

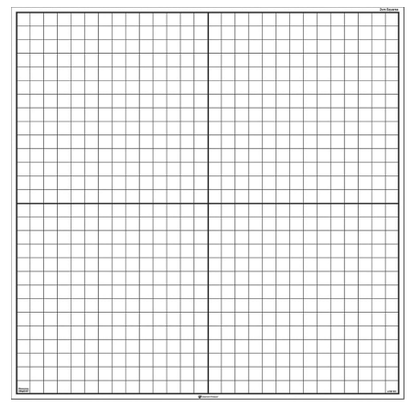
Grado y grupo _____ **Matricula** _____

1.- Grafique las siguientes funciones (20 puntos):

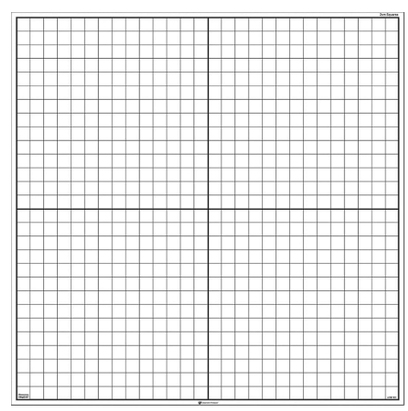
$y = 5x - 1$



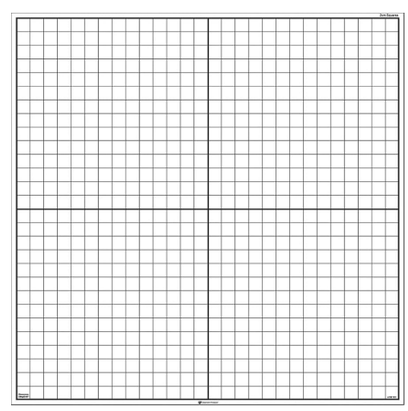
$y = \sqrt{x + 6}$



$2y - 2x = 0$

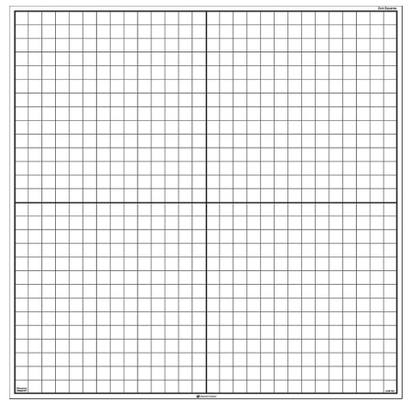


$y = x^2 + x + 3$

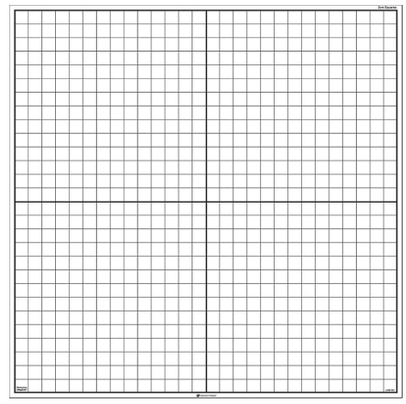


2.- Obtenga las siguientes asíntotas y grafiquémostelas (15 puntos)

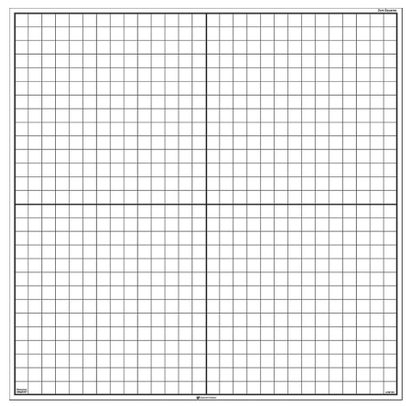
i) $$\frac{x-27}{x^2+4x+27}$$



ii) $$\frac{75x^2+4}{25x^2-4}$$

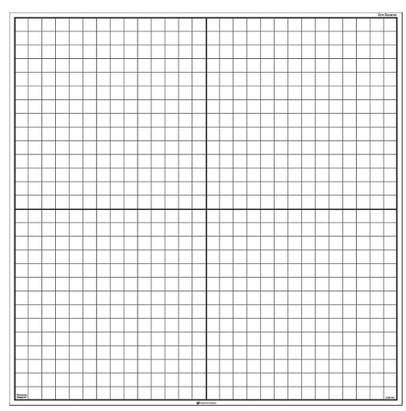


iii) $$\frac{x^2+5}{x-6}$$

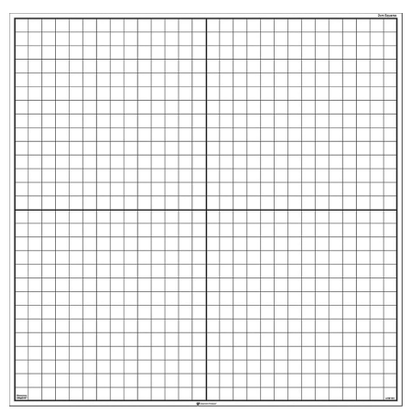


3.- Grafique las siguientes funciones e identifique si son creciente, decreciente o constante (15 puntos)

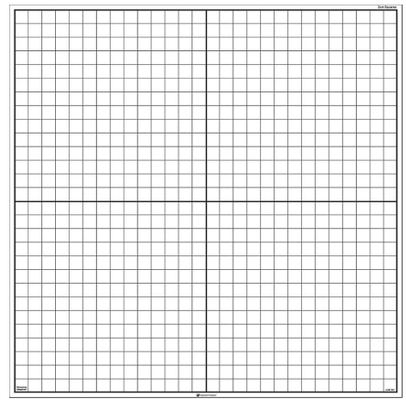
i) $f(x) = -3$



ii) $f(x) = 2x - 4$



iii) $f(x) = -x^2 + 5$



4.- Límites (20 puntos):

i) $\lim_{x \to 2} f(x) = x + 8$

ii) $\lim_{x \to 1} f(x) = 3x^3 + 2x^2 + x$

iii) $\lim_{x \to \infty} f(x) = -x^2 + 2x + 1$

iv) $\lim_{x \to -\infty} f(x) = x^3 - 900x^2 - 1234$

v) $\lim_{x \to 11} f(x) = \dfrac{x^2 - 64}{x - 8}$

vi) $\lim_{x \to 2} f(x) = \dfrac{x^2 - 4}{x - 2}$

vii) $\lim_{x \to 4} f(x) = \dfrac{x^2 - 1}{x^2 - 3x + 2}$

viii) $\lim_{x \to -1} f(x) = \dfrac{x^2 + 3x + 2}{x^2 - x - 2}$

ix) $\lim_{x \to \infty} f(x) = \dfrac{10x^2 + 2x + 123123}{2x^2 + 3}$

x) $\lim_{x \to \infty} f(x) = \dfrac{8x + 10}{4x^2 - 1}$

6.- Derivadas (20 puntos):

I) $f(x) = 20x^5 + 10x^4 + x^2 + 7x + 25876$

II) $f(x) = 2.76614389756109238$

III) $f(x) = \dfrac{10}{x^5} + \dfrac{20}{x^3} + \dfrac{30}{x^2}$

IV) $f(x) = \dfrac{1}{3}x^3 + \dfrac{1}{2}x^2 + \dfrac{5}{6}x + 256$

V) $f(x) = 12x^3 + \dfrac{2}{5}x$

VI) $f(x) = \sqrt[3]{7x^3}$

VII) $f(x) = \dfrac{2}{x^5} + \dfrac{3}{x^4} - \dfrac{4}{x^3}$

7.- Calcule la segunda y tercera derivada (10 puntos):

$y = 20x^5 + 20x^4 + 20x3$

$y = 10x^5 + 5x^4 - 2x^3 + 67$